

2023 Data Analysis Prelim Exam

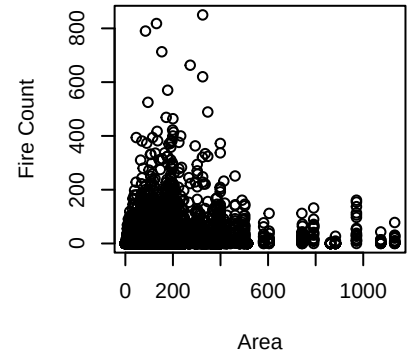
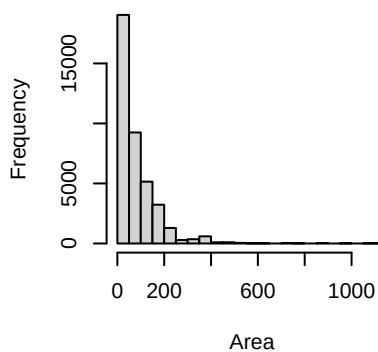
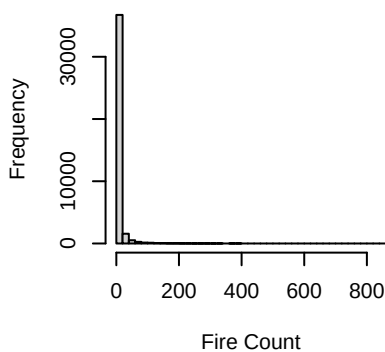
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Before any analysis, I load and clean the dataset. There are 55 variables and nearly 40,000 observations. There are a few columns that consist of mostly NA values. The missing values appear to be systematic, in most cases: for example, if a plantation never becomes certified, then an NA for `cert_year`, the year of certification, seems reasonable. Again, if a majority of the initial `f60_peat_for` values are zero, then it doesn't make sense to record a percentage change over time.

```
df <- read.csv("data.csv", header=TRUE)
```

Many of the variables appear skewed or have wide ranges. Others scale with the magnitude of the dataset, such as `f90_for`. For example, see the distribution of `fire_count`:

```
par(mfrow=c(1,3))
hist(df$fire_count,
     main="", xlab="Fire Count", breaks=40)
hist(df$total_area,
     main="", xlab="Area", breaks=40)
plot(df$total_area,
     df$fire_count, xlab="Area", ylab="Fire Count")
```



In analyses moving forward, I must carefully account for outliers, missing data, and erroneous data entry. With respect to the last point, I note that `pop_2000`, for example, contains a negative value. Since `pop_2000` represents the population density of the plantation in 2000, it's range should not include negative values. In answering the questions that follow, I will cautiously explore the data more with the context of each question in mind.

Q1

Q1a)

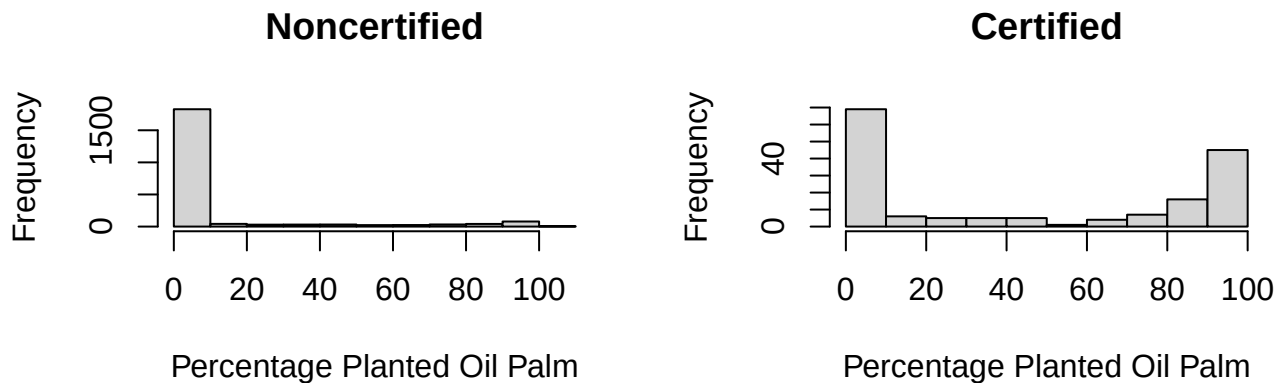
As stated in the question, Figure 2B depicts the percentage of the total plantation area that is planted with oil palm in the year 2000. In our dataset, `pp2000` represents the area of planted oil palm in 2000. We may

compute the percentage of oil palm opp as

$$\text{opp} = \frac{\text{pp2000}}{\text{area}} \cdot 100$$

. Differentiating between certified and noncertified plantations, I plot the data below.

```
df$opp <- df$pp2000/df$total_area*100
plantation <- df %>% filter(year==2000)
par(mfrow=c(1,2))
hist(plantation$opp[plantation$cert==0],
     xlab="Percentage Planted Oil Palm", main="Noncertified", breaks=12)
hist(plantation$opp[plantation$cert==1],
     xlab="Percentage Planted Oil Palm", main="Certified", breaks=12)
```



Notice that both groups contain a lot of mass near zero. However, the certified plantations is bimodal, with modes near 0 and 100. These plots agree with that of Figure 2B. I think that the histograms better depict the bimodal distribution in the certified case and the spike near zero in the non-certified case; the violin plots of Figure 2B appear to smooth over the spikiness of the modes present in each histogram.

Q1b)

The dataset includes a variable `f90_ptc1`, which measures the percentage change in `f90_for` from year to year. However, this is not exactly what we are interested in. We are interested in the percentage change in `f90_for` from 2001 to 2008. Instead, I compute the percentage change `pc` in `f90_for` from 2001 to 2008 for each plantation as

$$\text{pc} = \left| \frac{\text{f90for}(2008) - \text{f90for}(2001)}{\text{f90for}(2001)} \right| \cdot 100.$$

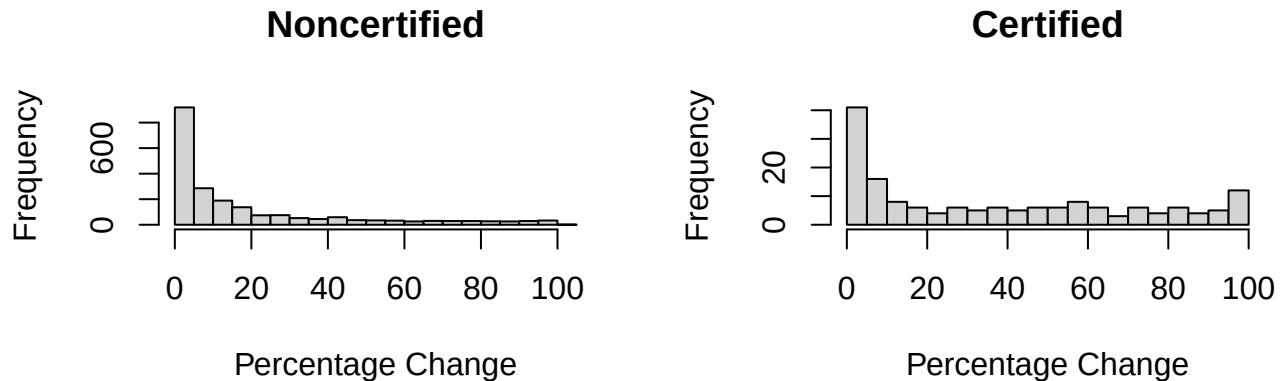
Note that the dataset is organized by ID and year, and so computation of `pc` is relatively straightforward.

```
init <- filter(df, year == 2001)
final <- filter(df, year == 2008)
plantation$pc <- abs(final$f90_for - init$f90_for)/init$f90_for*100
```

Some plantations have zero area of pixels with at least 90% tree cover (`'f90_for' = 0`). We may treat the corresponding `pc` values as 0, and include them in the histogram, or undefined, and exclude them from the histogram. I examine both: the distributions with and without the zero area values appear similar. For brevity, I display the distributions with the zero area values only.

```
par(mfrow=c(1,2))
# hist(plantation$pc[plantation$cert==0],
#      main="Noncertified", xlab="Percentage Change", breaks=20)
# hist(plantation$pc[plantation$cert==1],
```

```
#      main="Certified", xlab="Percentage Change", breaks=20)
plantation$pc[is.na(plantation$pc)] <- 0
hist(plantation$pc[plantation$cert==0],
     main="Noncertified", xlab="Percentage Change", breaks=20)
hist(plantation$pc[plantation$cert==1],
     main="Certified", xlab="Percentage Change", breaks=20)
```



Between certified and noncertified plantations, the two distributions appear more similar than they are depicted in Figure 2D. In particular, each histogram contains a mode near 0 and flattens out in the middle. As in part (a), the histograms depict a relative increase in mass near 100 for the certified plantations when compared to the non-certified plantations. The certified plantations appear to have a more uniform distribution, but the difference appears less pronounced than that shown in Figure 2D. We compute the entropy of the histogram to quantitatively compare the spread.

```
calc_entropy <- function(data) {
  probs <- table(data) / length(data)
  entropy <- -sum(probs * log2(probs))
  return(entropy)
}

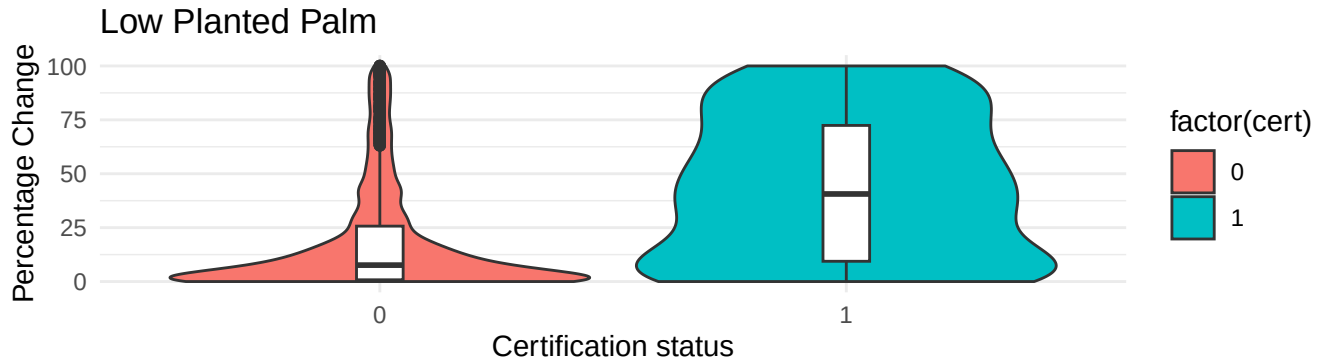
binned_0 <- cut(plantation$pc[plantation$cert==0],
               breaks=20, labels=FALSE)
binned_1 <- cut(plantation$pc[plantation$cert==1],
               breaks=20, labels=FALSE)
e0 <- calc_entropy(binned_0)
e1 <- calc_entropy(binned_1)
```

The certified group has a higher entropy distribution than the non-certified group ($3.8988 > 3.1329$), further suggesting more spread out change.

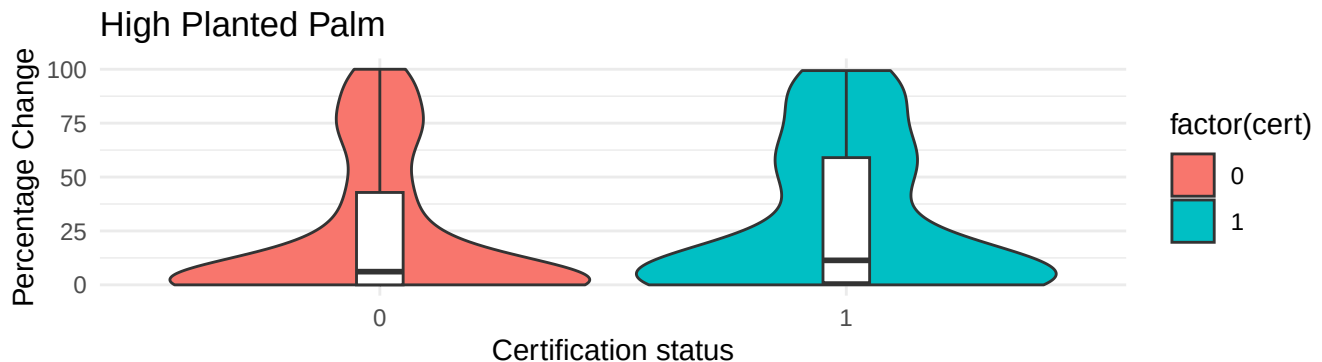
Q1c)

Figure 2B and my solution to **Q1(a)** demonstrate that the distribution of oil palm varies by certification group. In particular, certified plantations have more oil palms than non-certified plantations. The bimodal distribution corresponding to the certified group depicted in **Q1(a)**, suggests a mixture of plantations: a group of high palm oil certified plantations and low palm oil certified plantations. Meanwhile, the unimodal distribution corresponding to the non-certified group suggests only low palm oil non-certified plantations. It may not be fair to compare the overall deforestation between groups, as the underlying characteristics of the plantations differ. Splitting the plantations into subsets, those with more than 50% oil palm and those with no more than 50% oil palm, I attempt to recreate Figure 2D.

```
thres <- 50
subset0 <- plantation %>% filter(opp <= thres)
library(ggplot2)
ggplot(subset0, aes(x=factor(cert), y=pc, fill=factor(cert))) +
  geom_violin(scale="width", trim=TRUE) +
  geom_boxplot(width=0.1, fill="white") +
  theme_minimal() + xlab("Certification status") +
  ylab("Percentage Change")+ ggtitle("Low Planted Palm")
```



```
subset <- plantation %>% filter(opp > thres)
library(ggplot2)
ggplot(subset, aes(x=factor(cert), y=pc, fill=factor(cert))) +
  geom_violin(scale="width", trim=TRUE) +
  geom_boxplot(width=0.1, fill="white") +
  theme_minimal() + xlab("Certification status") +
  ylab("Percentage Change") + ggtitle("High Planted Palm")
```



By splitting the data into subsets based on the percentage of planted oil palm in 2000, I find that the deforestation patterns vary significantly by subset. For plantations with low levels of planted oil palm trees, the deforestation pattern appears to vary significantly between groups. Non-certified plantations experience low levels of deforestation, while the certified plantations experience more spread out levels of deforestation. On the other hand, for the plantations with moderate to high levels of planted oil palm trees, the distributions appear much more similar. Each group places most of its mass at low levels of deforestation.

Recall that in **Q1b**, we noticed that each group placed much mass near zero, but the certified plantations were more evenly distributed. The plots above help explain this phenomena. The first plot demonstrates differences between the groups, while the second plot illustrates similarity. For the differences, Consider the first plot, the low planted oil palm plot. The skew in the non-certified group helps explain the high spike near zero for the non-certified plantation's histogram in **Q1b** while the spread in the certified group is

consistent with the spread in the histogram of certified plantations in **Q1b**.

Q2

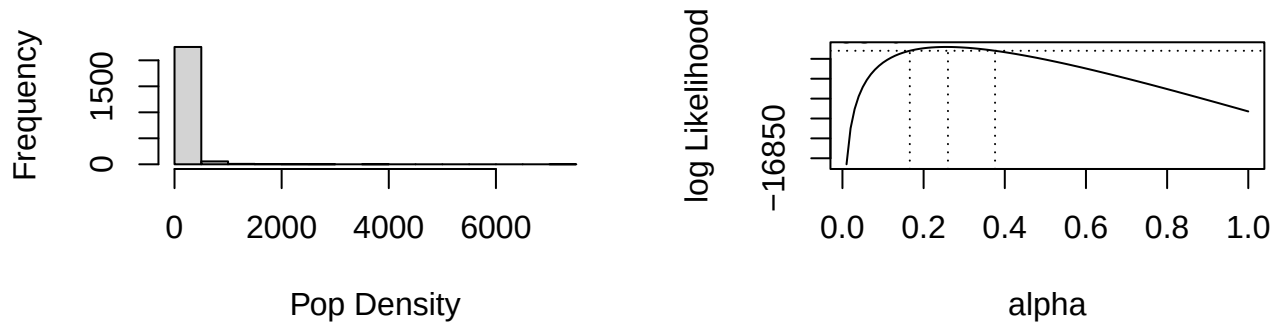
Our goal is to build a model for `pop_2000` based on the listed covariates. It is important to specify that the response variable is on the plantation level. Therefore, it is appropriate to use only one observation (year) per plantation. Since `t` and `p`, temperature and precipitation, vary by year, I consider their mean across years, by plantation, to obtain a plantation level covariate corresponding to each.

In **Q1**, I created such a dataframe called `plantation`, corresponding to the plantation characteristics. I use it in this problem, adding our plantation level temperature and precipitation covariates as columns to `plantation`.

```
grouped <- group_by(df, id) %>% summarise(mt = mean(t), mp = mean(p))
plantation$mt <- grouped$mt
plantation$mp <- grouped$mp
```

Recall that we found at least one negative `pop_2000` before answering **Q1**. Since a negative population density is incomprehensible, let's remove the negative values and view the distribution of the response variable, `pop_2000`.

```
par(mfrow=c(1,2))
positive <- filter(plantation, pop_2000 >= 0)
fit <- lm(pop_2000~dist_road + dist_port
          + elev + mt + mp + peat + total_area, data=positive)
hist(positive$pop_2000,
     main="", xlab="Pop Density")
tf <- logtrans(fit, alpha=seq(0.01, 1, by=0.01))
```



```
alpha <- tf$x[which.max(tf$y)]
positive$tf <- log(alpha + positive$pop_2000)
```

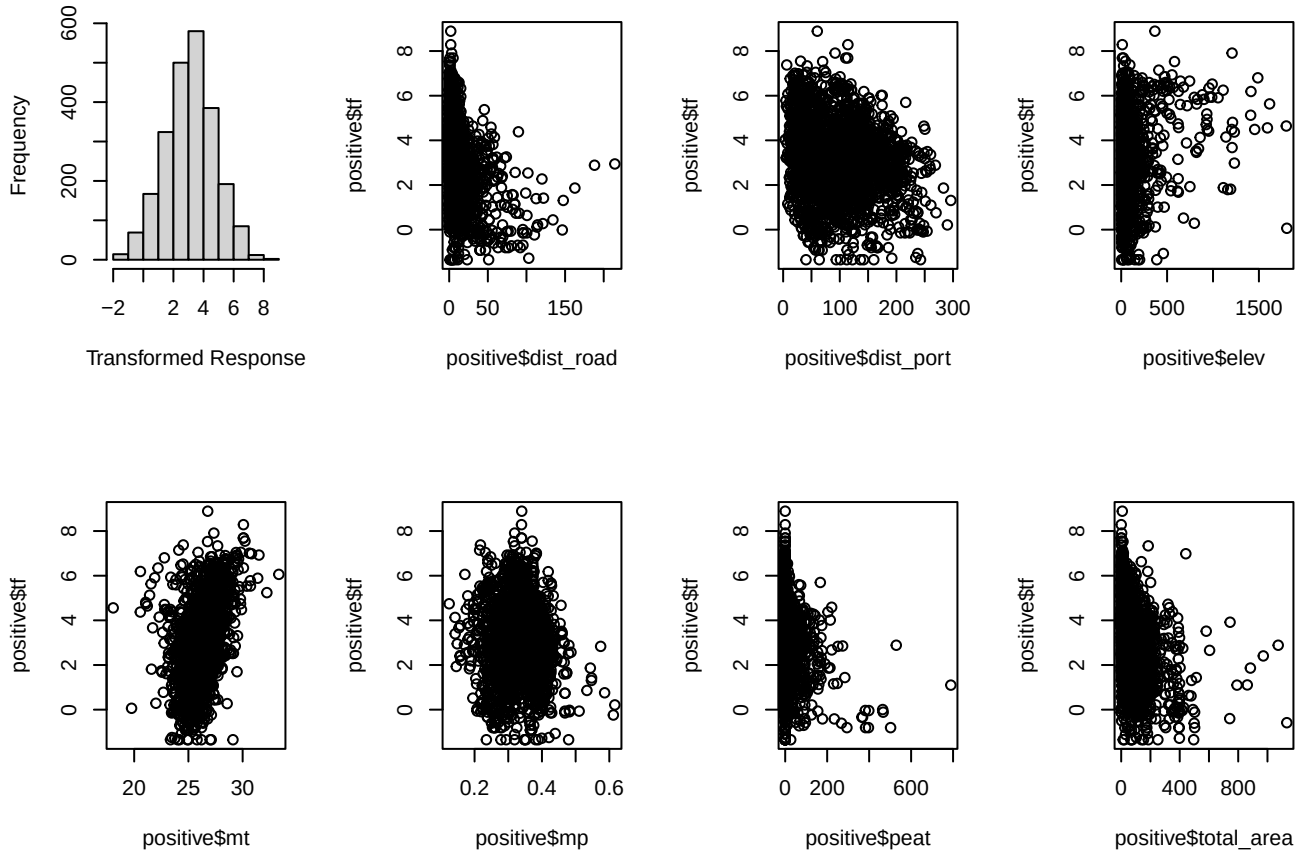
The response is highly skewed. A transformation appears appropriate. While a log-transform could seem natural, some of the response variables are zero. Instead, I try a shifted-log transform $f(x) : x \rightarrow \log(\alpha + x)$, $\alpha = 0.26$. The transformed response appears more symmetric and Gaussian. Any outliers, after transformation, do not appear to be outliers anymore. Plotting the relevant covariates against the transformed response, signs of heteroskedasticity are visible.

```
par(mfrow=c(2,4))
hist(positive$tf, xlab="Transformed Response",
     main="")
plot(positive$dist_road, positive$tf)
plot(positive$dist_port, positive$tf)
```

```

plot(positive$elev, positive$tf)
plot(positive$mt, positive$tf)
plot(positive$mp, positive$tf)
plot(positive$peat, positive$tf)
plot(positive$total_area, positive$tf)

```



In particular, the variance of `tf` decreases as most covariates, including `dist_road`, `peat`, `elev` and `total_area` increase. Some points that do not fit the overall trend of the data, either outliers or leverage points. Since there are only a few, I remove them. For example, see the rightmost point in the `peat` plot.

```

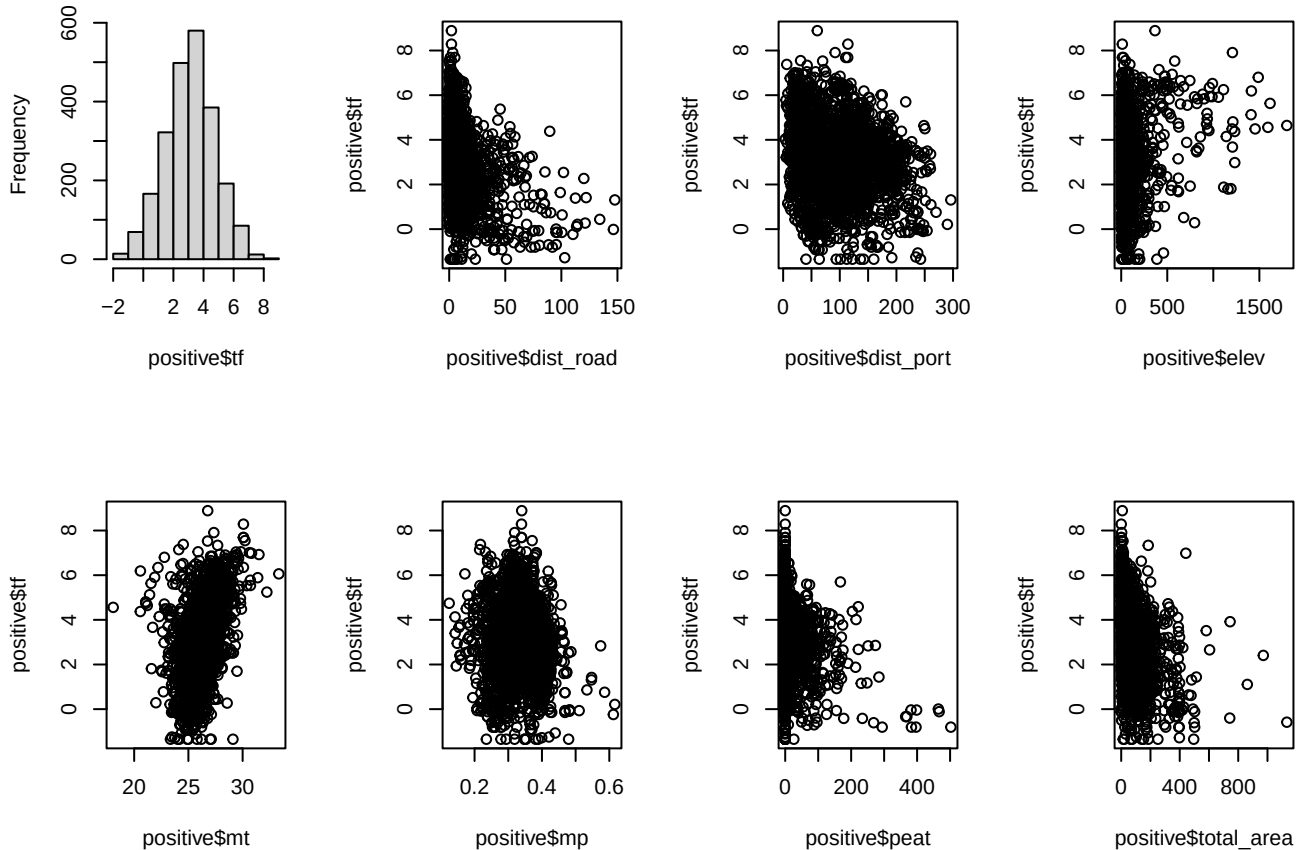
positive <- positive[-which.max(positive$peat),]
positive <- positive[-which.max(positive$dist_road),]
positive <- positive[-which.max(positive$dist_road),]
#remove second extraneous point
positive <- positive[-which.max(positive$dist_road),]
#remove third extraneous point
positive <- positive[-which.max(positive$elev),]

par(mfrow=c(2,4))
hist(positive$tf)
plot(positive$dist_road, positive$tf)
plot(positive$dist_port, positive$tf)
plot(positive$elev, positive$tf)
plot(positive$mt, positive$tf)
plot(positive$mp, positive$tf)
plot(positive$peat, positive$tf)

```

```
plot(positive$total_area, positive$tf)
```

Histogram of positive\$tf



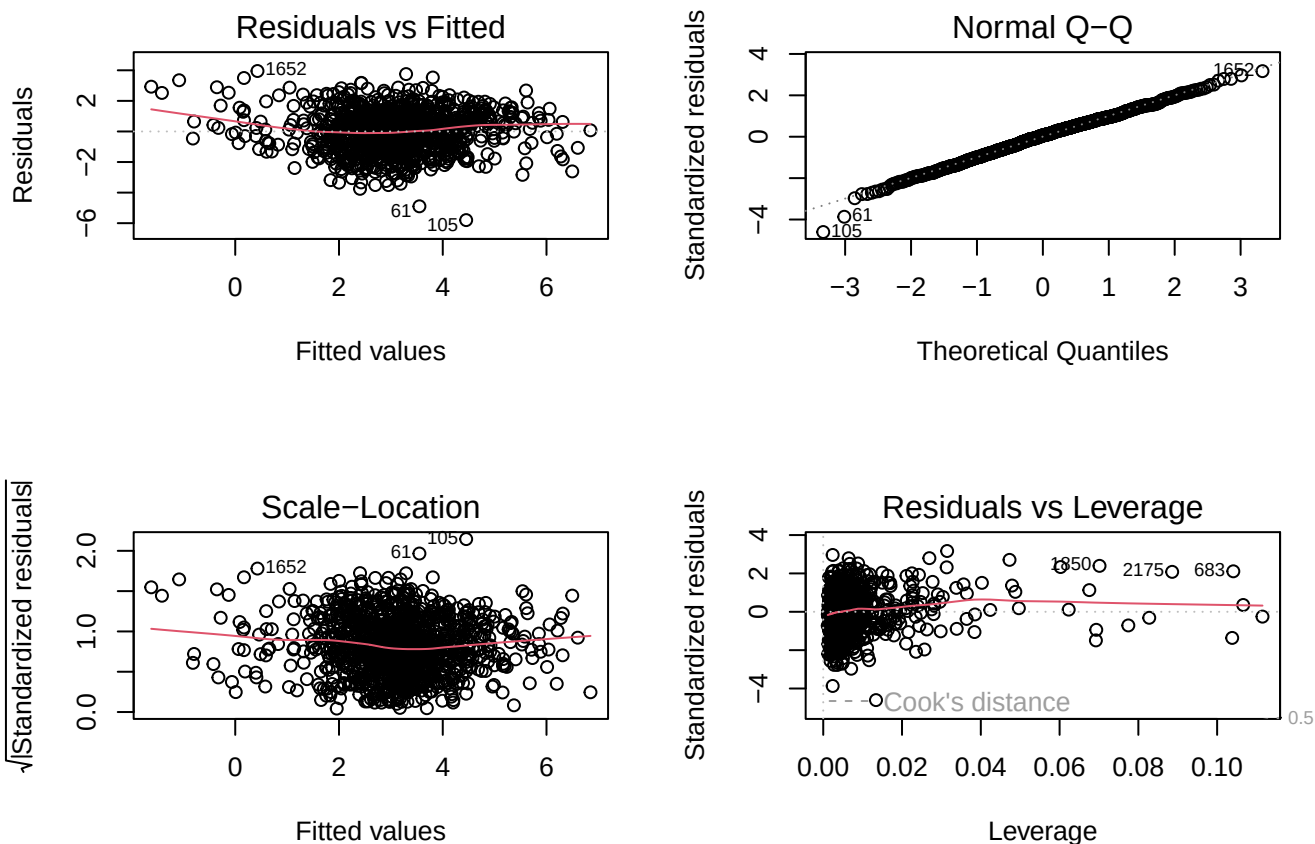
After removing outliers and points that do not fit the overall trend of the data, the diagnostic plots still show signs of nonconstant variance. I check for missing points (for brevity, I exclude this step from the analysis), and find 3 missing values in the `mt` column. I do not see a systematic reason for their missingness and there are very few compared to the size of the dataset. Therefore, I exclude them from the dataset and continue by splitting our data into train, validation, and test sets for model selection and fitting. Using the train set, I fit a baseline model with all of the specified covariates, excluding any interaction terms for now.

```
set.seed(123)
positive <- positive %>% subset(select=c(tf, dist_road,
                                         dist_port, mt, mp, elev, peat, total_area))
positive <- positive[complete.cases(positive),]
n <- nrow(positive)
inds <- sample(1:n, n/2)
train <- positive[inds,]
test <- positive[-inds,]
baseline <- lm(tf~dist_road + dist_port
               + elev + mt + mp + peat + total_area, data=train)
par(mfrow=c(2,2))
summary(baseline)

##
## Call:
## lm(formula = tf ~ dist_road + dist_port + elev + mt + mp + peat +
```

```
## total_area, data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.799 -0.843  0.076  0.861  3.954
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -13.793575   0.965666  -14.28  < 2e-16 ***
## dist_road    -0.014255   0.002554   -5.58 0.00000003 ***
## dist_port    -0.002272   0.000764   -2.97   0.003 **
## elev         0.003645   0.000268   13.62  < 2e-16 ***
## mt           0.635753   0.034656   18.34  < 2e-16 ***
## mp           1.549040   0.711738    2.18   0.030 *
## peat         -0.000772   0.001062   -0.73   0.467
## total_area   -0.002618   0.000495   -5.29 0.00000015 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.27 on 1153 degrees of freedom
## Multiple R-squared:  0.405, Adjusted R-squared:  0.402
## F-statistic: 112 on 7 and 1153 DF, p-value: <2e-16
```

```
plot(baseline)
```



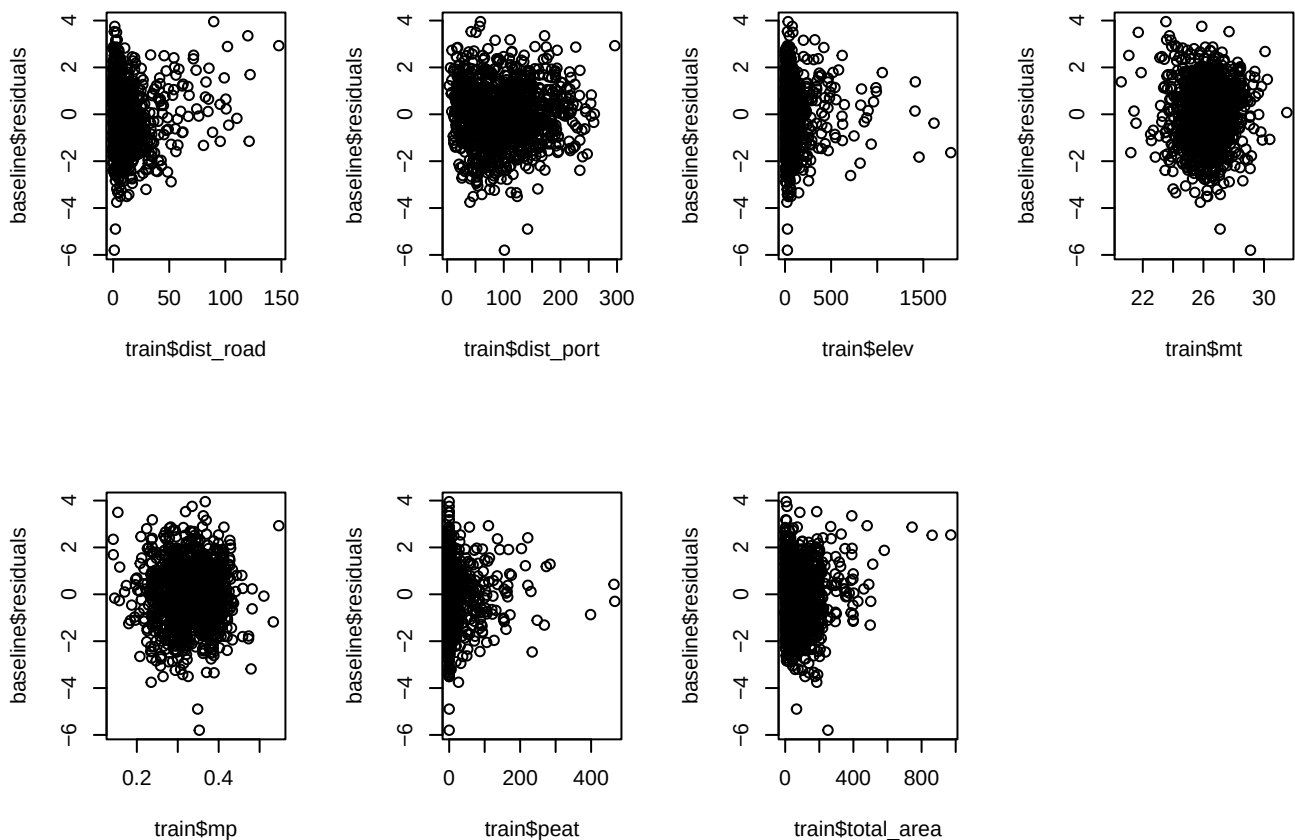
```
n=dim(train)[1];
degf=summary(baseline)$df[1]
```



```
msr <- max(abs(studres(baseline)))
bonf_thres <- qt(1-0.05/2/n,degf)
```

I check the studentized residuals for outliers and find none. The maximum studentized residual of 4.6451 is less than the Bonferroni-adjusted threshold of 8.0154. Diagnostic plots still show signs of nonconstant variance and nonlinearity.

```
par(mfrow=c(2,4))
plot(train$dist_road, baseline$residuals)
plot(train$dist_port, baseline$residuals)
plot(train$elev, baseline$residuals)
plot(train$mt, baseline$residuals)
plot(train$mp, baseline$residuals)
plot(train$peat, baseline$residuals)
plot(train$total_area, baseline$residuals)
```



A transformation of the covariates will leave us with the problem of nonconstant variance. Looking at the estimated coefficients, I find that `peat` is insignificant. Let's remove it from the model.

```
fit1 <- lm(tf~dist_road + dist_port +
  elev + mt + mp + total_area, data=train)
```

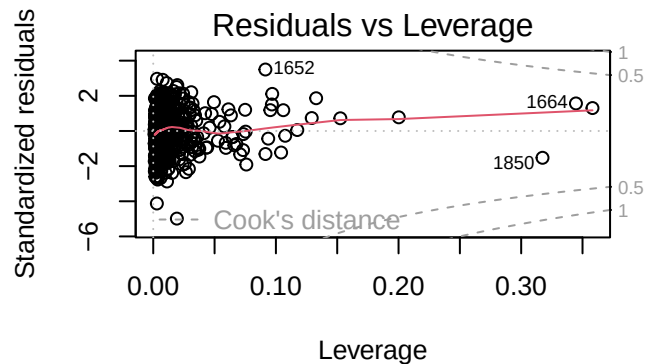
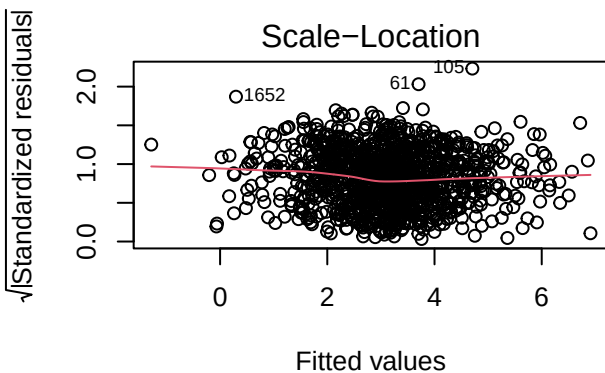
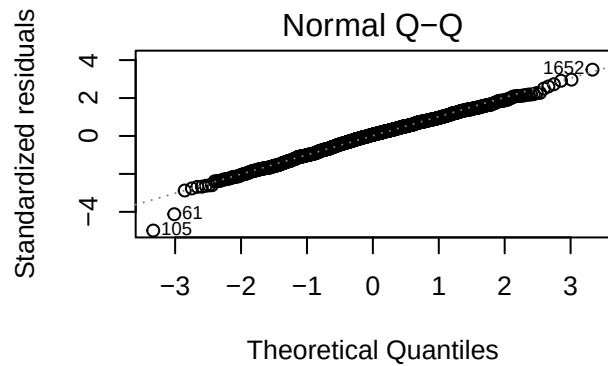
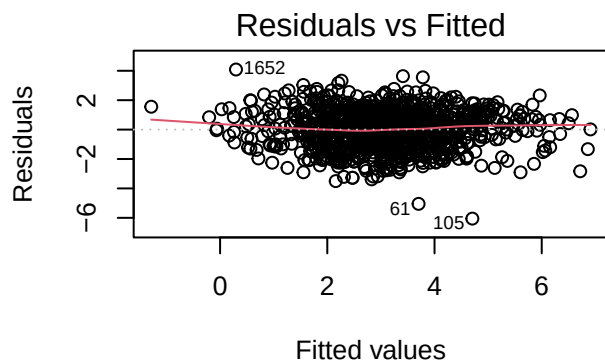
I check for interactions using backward stepwise regression with BIC as the selection criterion.

```
interact <- lm(tf~(dist_road + dist_port +
  elev + mt + mp + total_area)**2, data=train)
n <- nrow(train)
selected <- step(interact, k = log(n), direction="backward", trace=FALSE)
```

```
summary(selected)
```

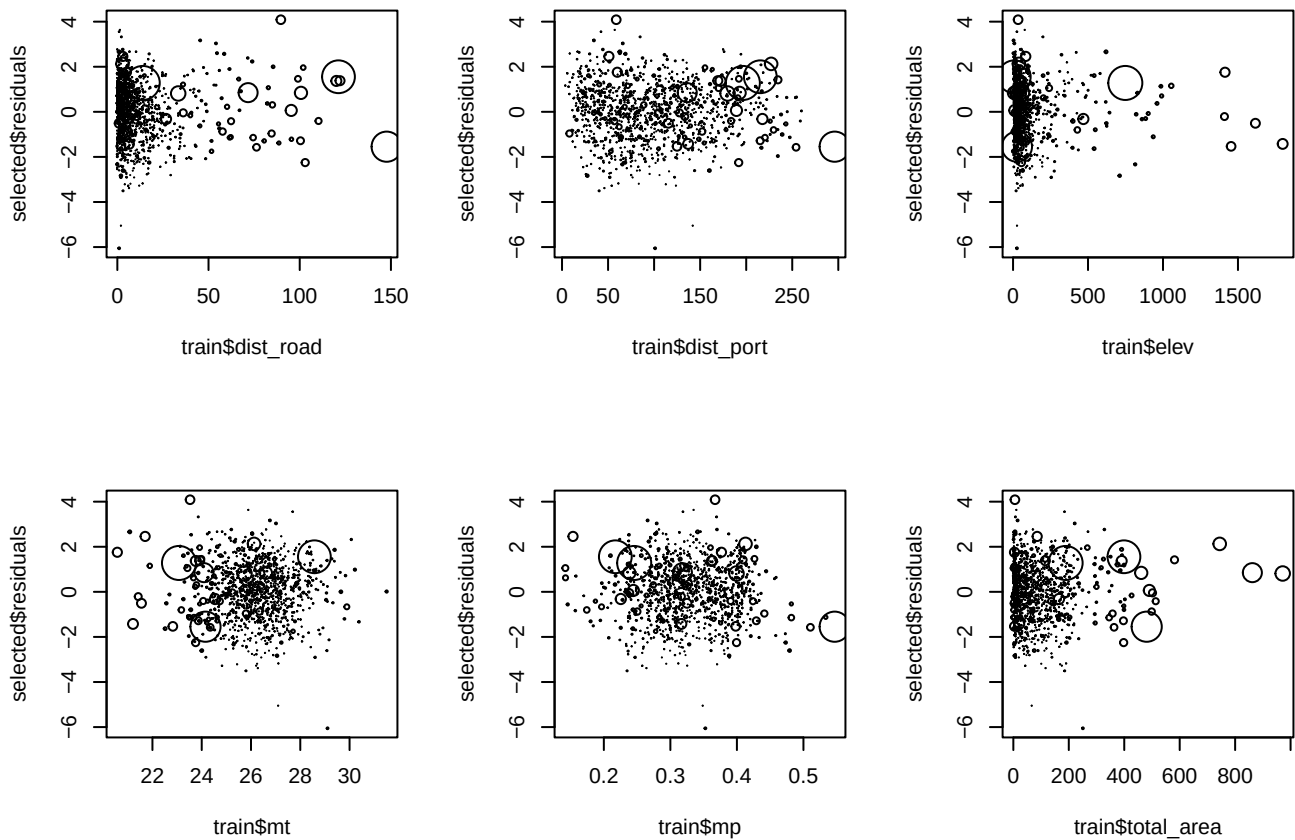
```
##
## Call:
## lm(formula = tf ~ dist_road + dist_port + elev + mt + mp + total_area +
##      dist_road:mt + dist_road:total_area + elev:total_area + mt:mp +
##      mp:total_area, data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.053 -0.815  0.079  0.846  4.086
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.31626129  4.17600489  -0.08  0.93964
## dist_road       0.14028906  0.04189215   3.35  0.00084 ***
## dist_port     -0.00209772  0.00075080  -2.79  0.00529 **
## elev           0.00385186  0.00028188  13.66 < 2e-16 ***
## mt             0.15176459  0.16002121   0.95  0.34312
## mp            -43.19598432  12.97444347  -3.33  0.00090 ***
## total_area     -0.01189281  0.00215188  -5.53 0.00000004 ***
## dist_road:mt   -0.00654745  0.00164203  -3.99 0.00007101 ***
## dist_road:total_area 0.00005302  0.00001370   3.87  0.00011 ***
## elev:total_area -0.00001860  0.00000608  -3.06  0.00225 **
## mt:mp          1.63444599  0.49742819   3.29  0.00105 **
## mp:total_area   0.02616633  0.00599888   4.36 0.00001405 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.23 on 1149 degrees of freedom
## Multiple R-squared:  0.446, Adjusted R-squared:  0.441
## F-statistic: 84.2 on 11 and 1149 DF, p-value: <2e-16

par(mfrow=c(2,2))
plot(selected)
```



The estimated coefficients lie above. I find that after correcting for multiple hypothesis testing concerns, using a Bonferroni correction at the $\frac{0.05}{(6)}_{(2)}$ significance level, we should include all listed interactions. Plotting the residuals and leverage points, The problem of nonconstant variance appears to have been largely resolved by including interaction terms in the model. Some points may be influential. I remove the largest five leverage points and refit the model accordingly.

```
X = model.matrix(selected)
H = (X%*%solve(t(X)%*%X,t(X)))
lev = diag(H)
par(mfrow=c(2,3))
plot(train$dist_road, selected$residuals, cex=10*lev)
plot(train$dist_port, selected$residuals, cex=10*lev)
plot(train$elev, selected$residuals, cex=10*lev)
plot(train$mt, selected$residuals, cex=10*lev)
plot(train$mp, selected$residuals, cex=10*lev)
plot(train$total_area, selected$residuals, cex=10*lev)
```



```
top5 = order(diag(H))[length(diag(H)):(length(diag(H))-4)]
selected_rev <- lm(tf ~ dist_road + dist_port + elev + mt + mp + total_area +
  dist_road:mt + dist_road:total_area + elev:total_area + mt:mp +
  mp:total_area, data = train[-top5, ])
n=dim(train)[1];
degf=summary(baseline)$df[1]
msr <- max(abs(studres(selected)))
bonf_thres <- qt(1-0.05/2/n,degf)
c <- cor(fitted(selected)[-top5], fitted(selected_rev))
```

I check the studentized residuals for outliers and find none. The maximum studentized residual of 5.0384 is less than the Bonferroni-adjusted threshold of 8.0154. The model's fitted values with and without the leverage points have a correlation of 0.999, demonstrating that the leverage points minimally impact OLS estimates. Signs of heteroskedasticity have largely disappeared. It is hard to tell whether the variance appears to decrease slightly with `total_area`, however if such a decrease is present, it is minimal, and so we do not think the assumption of homoskedasticity is violated severely. A careful approach may consider bootstrapping to compare the OLS-estimated standard errors with bootstrapped standard errors.

Let's fit this model to the test data to estimate our coefficients and derive a final model.

```
test_fit <- lm(tf ~ dist_road + dist_port + elev + mt + mp + total_area +
  dist_road:mt + dist_road:total_area + elev:total_area + mt:mp +
  mp:total_area, data=test)
pvals <- summary(test_fit)$coefficients[,4]
```

I find that `dist_port` has an estimated p-value of 0.1064 and so we conclude that the effect of `dist_port`, after controlling for the other included covariates, does not have an effect on the population density.

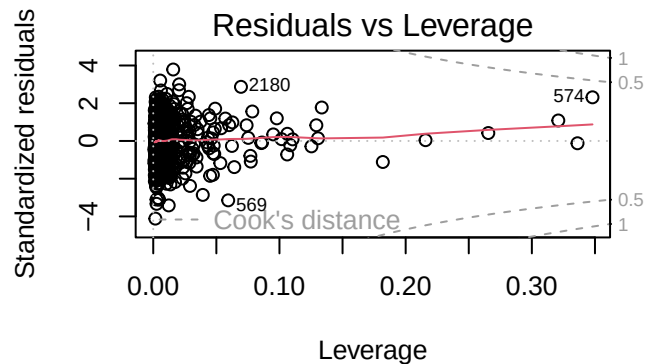
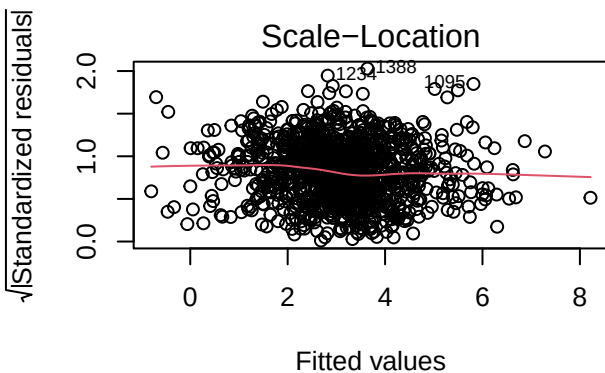
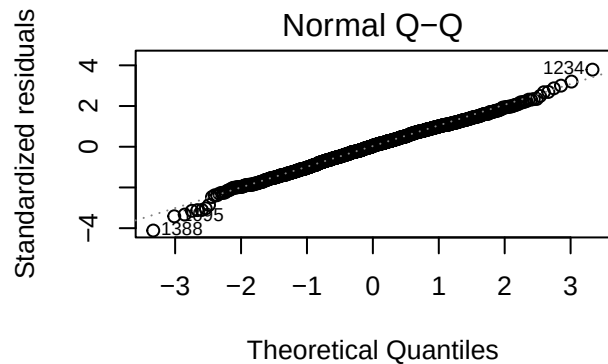
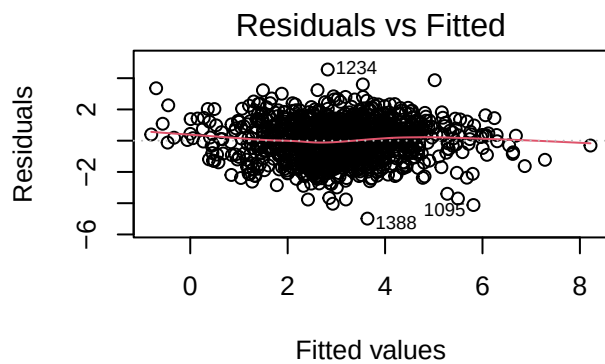
```

formula <- tf ~ dist_road + elev + mt + mp + total_area +
  dist_road:mt + dist_road:total_area + elev:total_area + mt:mp +
  mp:total_area
final <- lm(formula, data=test)
summary(final)

##
## Call:
## lm(formula = formula, data = test)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.985 -0.786  0.044  0.870  4.555
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -5.14459975   3.73507220   -1.38    0.1687
## dist_road      0.07197626   0.05099561    1.41    0.1584
## elev          0.00341553   0.00026450   12.91 < 2e-16 ***
## mt            0.33048236   0.14154549    2.33    0.0197 *
## mp           -29.45198724  11.39753188   -2.58    0.0099 **
## total_area    -0.00974094   0.00216187   -4.51 0.00000729 ***
## dist_road:mt  -0.00419639   0.00195809   -2.14    0.0323 *
## dist_road:total_area 0.00007576 0.00001503    5.04 0.00000053 ***
## elev:total_area -0.00001281 0.00000297   -4.31 0.00001797 ***
## mt:mp          1.12252718   0.43315413    2.59    0.0097 **
## mp:total_area   0.01572497   0.00604952    2.60    0.0095 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.21 on 1150 degrees of freedom
## Multiple R-squared:  0.471, Adjusted R-squared:  0.466
## F-statistic: 102 on 10 and 1150 DF, p-value: <2e-16

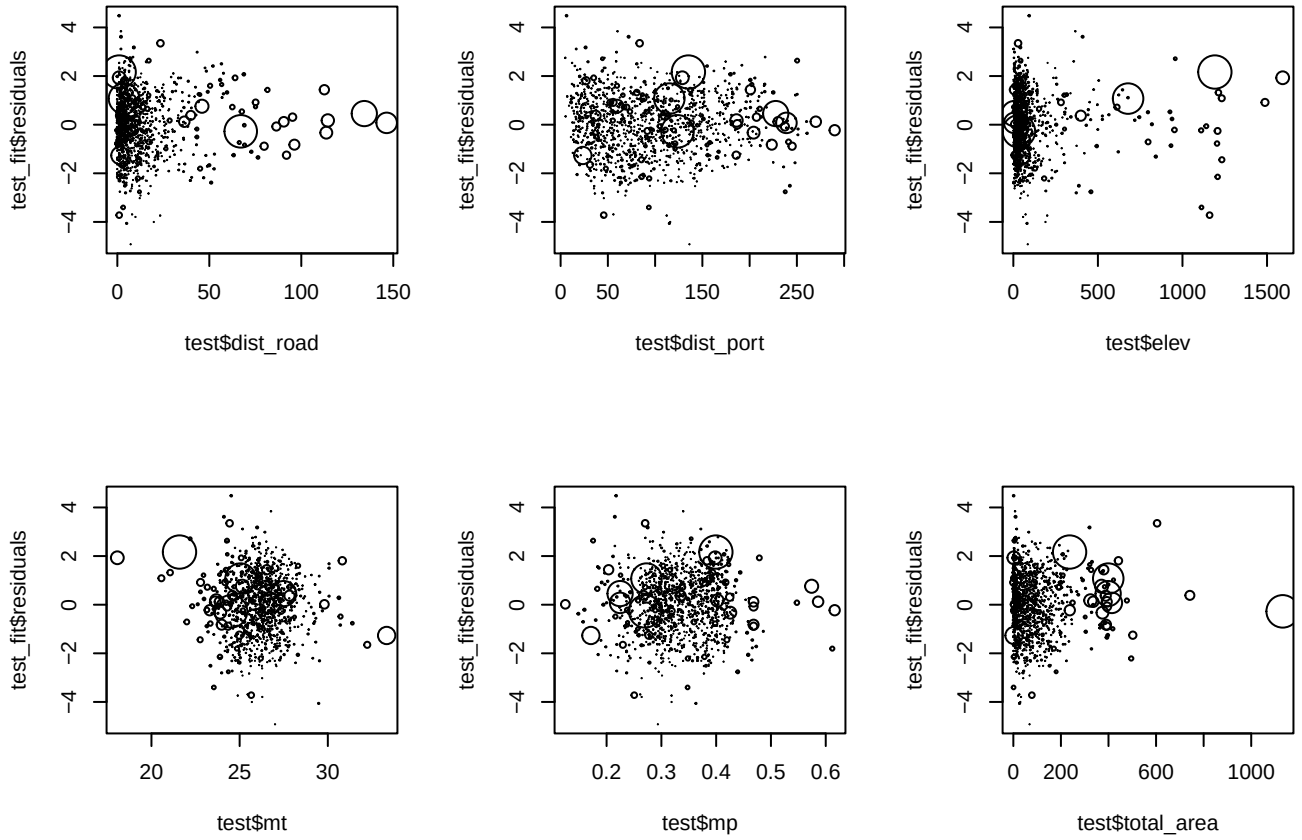
pvals <- summary(test_fit)$coefficients[,4]
par(mfrow=c(2,2))
plot(final)

```



Our final estimated coefficients lie above. Note that the effect of distance to nearest road on the population density is decreasing with temperature and increases with total area. The intercept effect of precipitation is negative but increases with area and temperature. Warm, wet conditions may be suitable for agriculture and life to flourish.

```
X = model.matrix(test_fit)
H = (X%*%solve(t(X)%*%X,t(X)))
lev = diag(H)
par(mfrow=c(2,3))
plot(test$dist_road, test_fit$residuals, cex=10*lev)
plot(test$dist_port, test_fit$residuals, cex=10*lev)
plot(test$elev, test_fit$residuals, cex=10*lev)
plot(test$mt, test_fit$residuals, cex=10*lev)
plot(test$mp, test_fit$residuals, cex=10*lev)
plot(test$total_area, test_fit$residuals, cex=10*lev)
```



We must carefully consider the effect of `total_area` on the transformed population density since it is implicitly in the denominator of the population density. However, The residual plots look okay and none of the leverage points look too problematic. A transformation of some covariates may be meaningful in future analyses. Every variable included in the model is significant at the $p = 0.05$ level and so we include all terms in our final model. While `dist_road` has a p-value of 0.1996, its corresponding interactions are significant, and so we include the intercept term in our final model. Our final model for plantation i is:

$$\log(\text{pop.2000}_i + \alpha) = X_i\beta + \sigma\epsilon_i; \quad \epsilon_i \sim N(0, 1)$$

Q3

We are now interested in the probability that a plantation decides to start the certification initiation process based on both the plantation's characteristics and past year's information. Here, we must be careful: some covariates **completely determine** the `certintent_now` variable. For example, `certintent_now = 0` for all years before 2009. A plantation must be a member of the RSPO before they can be certified. Once a plantation has started the certification initiation, `certintent_now == 1` for all future years. Therefore, I first trim the dataframe to the subset of observations in which a plantation **could** decide to start the certification initiation process in the first place. Since no plantations start initiation before 2009, we remove all observations before 2008.

```
df$certintent_year[is.na(df$certintent_year)] <- 9999
df$reduct <- (df$f90_for - c(df$f90_for[2:nrow(df)], 0))
df$reduct[is.na(df$reduct)==1] <- 0
subset_df <- df %>% filter(pop_2000 >= 0 &
  year < certintent_year & year >= rspo_year)
```

```
& year >= min(certintent_year) - 1)
subset_df$cert_next <- subset_df$certintent_year == (subset_df$year + 1)
```

Note that `cert_next` is an indicator for whether or not a plantation starts the initiation process in the next year. Therefore, we may treat `cert_next` as the response variable, as `cert_next` directly corresponds to `certintent_now` in the following year. However, now each row of `subset_df` includes the response variable, the plantation characteristics, and the information of the year before `certintent_now`, which is what we desire. Therefore, we may use `cert_next` as our response variable and the columns of `subset` for our regression analysis. I remove columns that contain many NA values or that should be excluded from the model.

```
subset_df <- subset_df %>% subset(select=-c(f90_peat_ptcl,
  f90_prim_ptcl, f60_peat_ptcl, f60_prim_ptcl, f30_prim_ptcl,
  f30_peat_ptcl, country, province, district, island, admin,
  cert_year, cert, random))
subset_df <- subset_df[complete.cases(subset_df),]
up <- length(unique(subset_df$id))
```

There are only 370 plantations in our subset, 154 of which that commence initiation at all, and 2059 year-plantation observations. I create a baseline model including many covariates, most of which are not significant. As in **Q2**, I split our data into train data for model selection and test data for model fitting.

```
n <- nrow(subset_df)
inds <- sample(1:n, n/2)
train3 <- subset_df[inds, ]
test3 <- subset_df[-inds, ]
baseline <- glm(cert_next~t + p + pop_2000 +
  dist_port + dist_pp + dist_road + total_area
  + elev + slope + peat + fire_rate + f30_for + f30_ptcl
+ f60_for + f60_ptcl + f90_for + f90_ptcl + fire_count
+ fire_rate + factor(year) + pp2000 + pp2005 + pp2010 +
  otp2000 + otp2005 + otp2010, family=binomial(), data=train3)
```

Let's use backward stepwise regression with AIC as the selection criteria to reduce the number of covariates in the model. With many covariates, AIC tends to overestimate the number of true nonzero coefficients. Therefore, we use backward stepwise regression with p-value as the selection criteria to reduce the number of covariates further.

```
fit <- step(baseline, direction="backward", trace=FALSE)
fit1 <- glm(cert_next ~ p + elev + peat + f30_ptcl + f60_for +
  f90_for + factor(year) + pp2000 + pp2010, family = binomial(),
  data = train3)
fit2 <- glm(cert_next ~ p + elev + peat + f30_ptcl + f60_for +
  f90_for + factor(year) + pp2010, family = binomial(),
  data = train3)
fit3 <- glm(cert_next ~ p + elev + peat + f30_ptcl +
  f90_for + factor(year) + pp2010, family = binomial(),
  data = train3)
fit4 <- glm(cert_next ~ p + elev + f30_ptcl +
  f90_for + factor(year) + pp2010, family = binomial(),
  data = train3)
fit5 <- glm(cert_next ~ p + f30_ptcl +
```



```
f90_for + factor(year) + pp2010, family = binomial(),
data = train3)
d5 <- fit5$deviance
```

Should we include a random effect of year in our model? Decisions to start the initiation process may be correlated by year.

```
train3$year <- factor(train3$year)
fit_re <- glmer(cert_next ~ f30_ptcl
+ f90_for + pp2010 + p + (1|fyear), data=train3, family=binomial())
cmp <- anova(fit_re, fit5, test="LRT")
```

We execute a likelihood ratio test and determine that we should keep the year effect a fixed effect, obtaining a p-value of 0.002.

```
interact <- glm(cert_next ~ (p + f30_ptcl +
f90_for + factor(year) + pp2010)**2, family=binomial(), data=train3)
interact2 <- glm(cert_next ~ (p + f30_ptcl +
f90_for + pp2010)**2 + factor(year), family=binomial(), data=train3)
cmp1 <- anova(fit5, interact, test="LRT")[2,5]
cmp2 <- anova(fit5, interact2, test="LRT")[2,5]
```

I also check for interaction terms. I use a likelihood ratio test to see if I should include any interaction terms and find that I should not ($\hat{p} = 0.4958, 0.6727$).

```
cloglog <- glm(cert_next ~ f30_ptcl + pp2010
+ p + f90_for + factor(year), family=binomial(link="cloglog"),
data=train3)
d <- cloglog$deviance
```

I also check for asymmetry in the response using a cloglog link function, but the residual deviance is higher (417.1978 vs. 416.5791), so we do not use this link function. We stick with the logit link function. Thus, our final model is fit5. I estimate the model's coefficients using our test data. Note that the only plantation characteristic is pp_2010, a measurement taken a year after certification initiation became available.

```
final <- glm(cert_next ~ f30_ptcl +
pp2010 + p + f90_for + factor(year),
family=binomial(), data=test3)

final2 <- glm(formula =
cert_next ~ f30_ptcl + pp2010 + f90_for + factor(year),
family = binomial(), data = test3)
comp <- anova(final2, final, test="LRT")[2,5]
summary(final2)
```

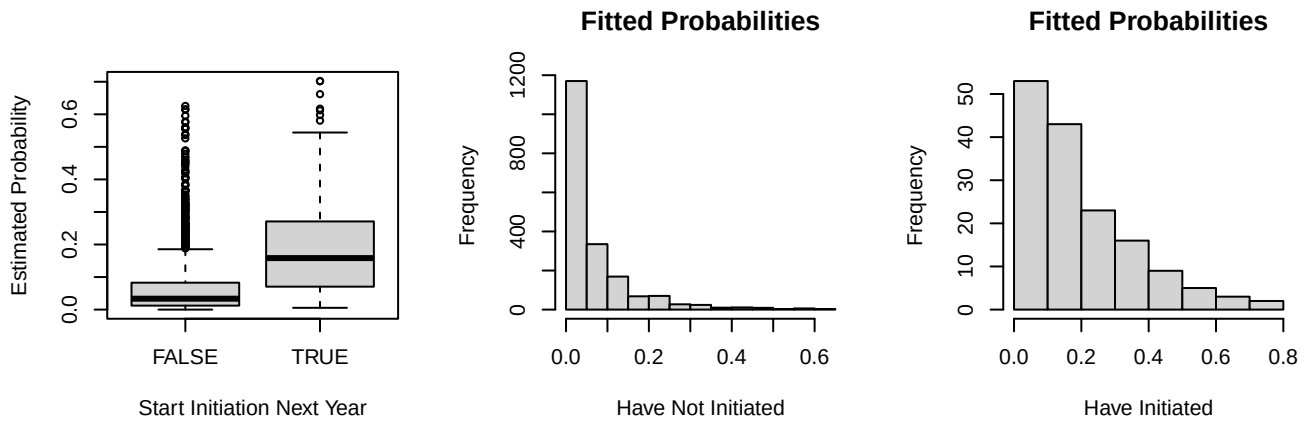
```
##
## Call:
## glm(formula = cert_next ~ f30_ptcl + pp2010 + f90_for + factor(year),
##      family = binomial(), data = test3)
##
## Deviance Residuals:
##      Min        1Q    Median        3Q        Max
## -1.383   -0.387   -0.231   -0.121    3.225
```

```
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.48641    0.48940   -7.12  1.0e-12 ***
## f30_ptcl      -6.65425    2.42855   -2.74  0.00614 **
## pp2010         0.01541    0.00232    6.64  3.1e-11 ***
## f90_for       -0.04184    0.01521   -2.75  0.00595 **
## factor(year)2009  0.85710    0.53490    1.60  0.10908
## factor(year)2010  1.33107    0.51324    2.59  0.00950 **
## factor(year)2011 -0.30020    0.67881   -0.44  0.65831
## factor(year)2012  1.92034    0.51264    3.75  0.00018 ***
## factor(year)2013  0.38345    0.61738    0.62  0.53454
## factor(year)2014  0.62277    0.61945    1.01  0.31472
## factor(year)2015 -0.61114    0.84967   -0.72  0.47197
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 552.51  on 1029  degrees of freedom
## Residual deviance: 433.95  on 1019  degrees of freedom
## AIC: 455.9
##
## Number of Fisher Scoring iterations: 8
```

A likelihood ratio test suggests that we remove p from the model, yielding a p-value of 0.0171. Thus, my final model for plantation i in year j is:

$$(\text{logit}(P(\text{certintent.now}_{ij} = 1 \mid X))) = \beta_0 \text{f30.ptcl}_{ij-1} + \beta_1 \text{pp2010}_i + \beta_2 \text{f90.for}_{ij-1} + \beta_j 1_{\text{year}=j-1}$$

```
fits <- exp(predict.glm(final2, newdata=subset_df))/(1 +
              exp(predict.glm(final2, newdata=subset_df)))
par(mfrow=c(1,3))
plot(factor(subset_df$cert_next), fits,
      xlab="Start Initiation Next Year", ylab="Estimated Probability")
hist(fits[subset_df$cert_next==0],
      xlab="Have Not Initiated", main="Fitted Probabilities")
hist(fits[subset_df$cert_next==1],
      xlab="Have Initiated", main="Fitted Probabilities")
```



Note that in both the baseline model and the final logistic regression model, we see that the estimated probabilities for those that have initiated tend to be larger than the plantations that have not initiated the certification process. Here, I interpret the two groups to be the groups separated by the response: the group that have not initiated corresponds to `cert_next=0` (`certintent_now=0`), while the group that has initiated corresponds to `cert_next=1` (`certintent_now=1`) in `subset_df`.

Q4

Q4(a)

Please see the final two pages of this document for derivation of the EM algorithm.

Q4(b)

Implementation is relatively straightforward: we alternate between updating $\{\tau_{ik}\}$ and updating $\{\hat{\mu}_k^{(t)}\}$ until convergence. I first implement the E-step, updating τ .

```
update_tau <- function(MU, Y, x){
  Z <- matrix(0, N, K)
  for (i in 1:N){
    llk <- rep(0, K)
    c <- x[i]
    llk <- - c*rowSums(MU) + log(c*MU) %*% Y[i,] - log(K)
    probs <- exp(llk - logsumexp(llk))
    Z[i,] <- probs
  }
  return(Z)
}
```

I then implement the M step, updating $\Theta = \{\hat{\mu}_k^{(t)}\}_{k,t}$.

```
update_MU <- function(Y, x, Z, a, b){
  K <- dim(Z)[2]
  TT <- dim(Y)[2]
  alpha <- t(Z) %*% Y
  beta <- t(Z) %*% x
  MU <- matrix(0, nrow=K, ncol=TT)
  for (k in 1:K){
    for (t in 1:TT){
      MU[k,t] <- (alpha[k,t] + a - 1)/(b + beta[k])
    }
  }
}
```

```

}
}
return(MU)
}

```

Q4c)

To pre-process the data, I create the specified plantation $\times$ year matrix $Y \in \mathbb{R}^{N \times T}$ and **total_area** vector $x \in \mathbb{R}^N$.

```

N <- 2331
TT <- 17
Y <- matrix(0, nrow=N, ncol=TT)
for (i in 1:nrow(df)){
  id <- df[i,1]
  t <- df[i,2] - 1999
  Y[id, t] <- df[i, 54]
}

x <- rep(0, N)
for (i in 1:N){
  a <- plantation$total_area[which(plantation$id == i)]
  x[i] <- a
}

```

Q4d)

As specified, I set $K = 7$, $a_0 = 1.15$, and $b_0 = 0.5$ and fit EM, initializing $\mu_k^{(t)}$ randomly sampled i.i.d. $\sim \Gamma(0.01, 0.01)$. Such an initialization scheme has high variance and mean of 1.

```

set.seed(456)
K <- 7
a <- 1.15
b <- 0.5
diff <- 1000
iter <- 1

#initialize mu
MU <- matrix(0, K, TT)
for (k in 1:K){
  MU[k,] <- rgamma(TT, 0.01, scale=0.01)
}
Z_old <- matrix(0, nrow=N, ncol=K)
MU_old <- MU
while (diff > 0.001){
  Z <- update_tau(MU, Y, x)
  MU <- update_MU(Y, x, Z, a, b)
  diff <- sum(abs(Z - Z_old)) + sum(abs(MU - MU_old))
  Z_old <- Z
  MU_old <- MU
}

```

```

if (iter %% 10 == 0){
  print(paste0("iteration: ", iter))
  print(diff)
}
iter <- iter+ 1
}

```

```

## [1] "iteration: 10"
## [1] 239.83
## [1] "iteration: 20"
## [1] 35.964
## [1] "iteration: 30"
## [1] 1.9952
## [1] "iteration: 40"
## [1] 0.68306
## [1] "iteration: 50"
## [1] 0.014782
## [1] "iteration: 60"
## [1] 0.00083941

```

Q4e)

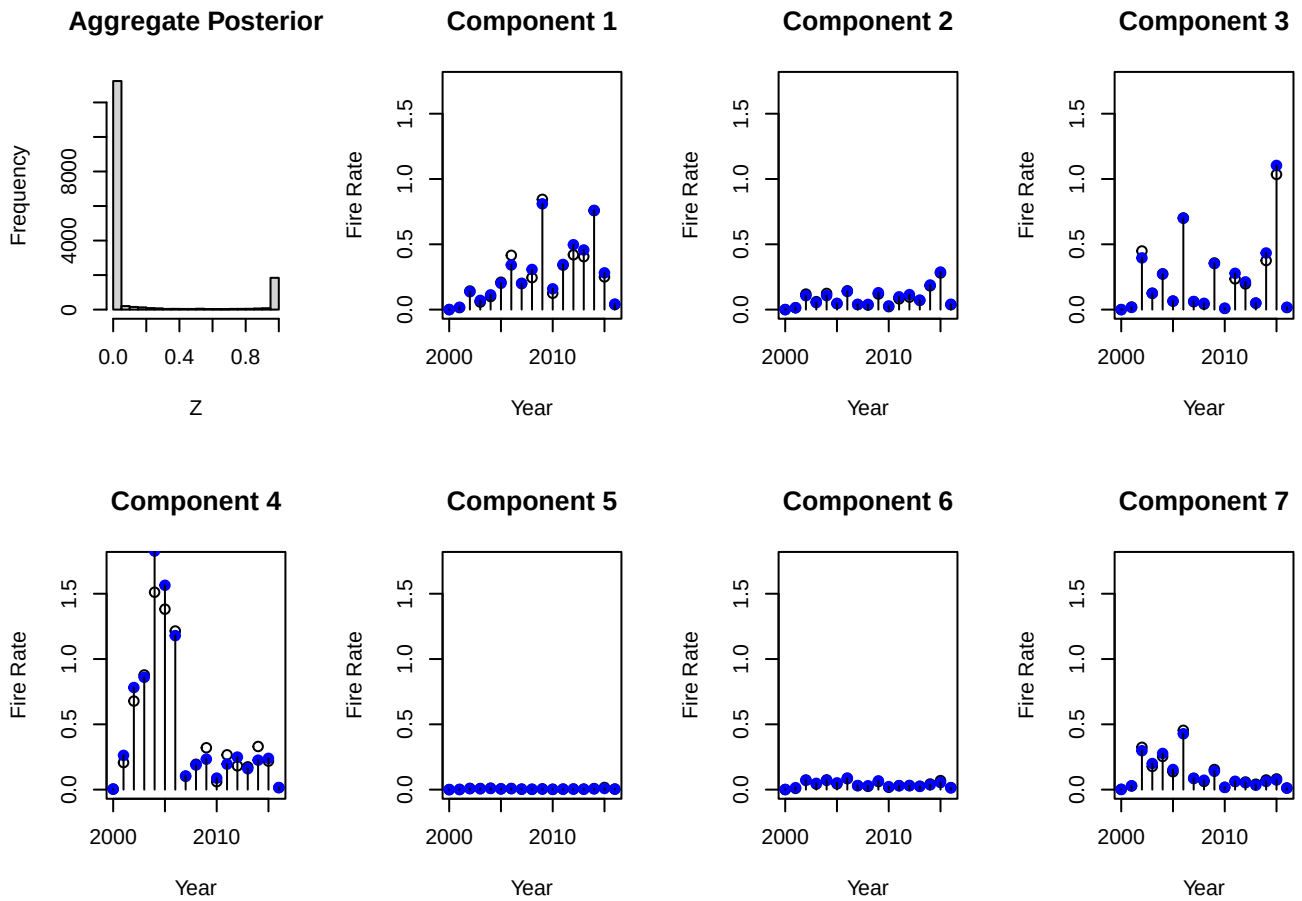
Upon convergence, let's look at a few of the latent components. The histogram of expected latent assignments is low-entropy, with most of the plantations having a high probability of matching one component. I also plot the mean `fire_rate` by component, first assigning each plantation i to $\arg \max_k \tau_{ik}$, and then averaging over fire rate by year for each component. The plots are shown below.

```

maxes <- rep(0, N)
for (i in 1:N){
  maxes[i] <- which.max(Z[i, ])
}

par(mfrow=c(2,4))
hist(Z, main="Aggregate Posterior")
for (k in 1:K){
  group <- which(maxes==k)
  group_df <- df %>% filter(id %in% group)
  gdf <- group_df %>% group_by(year)
  ym <- summarise(gdf, fr=mean(fire_rate))
  plot(ym$year, MU[k,], main=paste0("Component ",k),
       xlab="Year", ylab="Fire Rate", ylim=c(0, 1.75))
  points(ym$year, ym$fr, col="blue", pch=19)
  segments(x0=ym$year, x1=ym$year, y0=0, y1=ym$fr)
}

```



The estimated $\mu_k^{(t)}$ match the empirical fire_rate data. Moreover, the plots depict heterogeneity across components. Notice how components 1 captures high fire rates in 2009 while component 4 captures high fire rates in the early 2000s. Component 3 appears to capture higher fire rates in 2006, 2015, while components 2, 5, and 6 correspond to plantations with uniformly low fire rates.

Q5

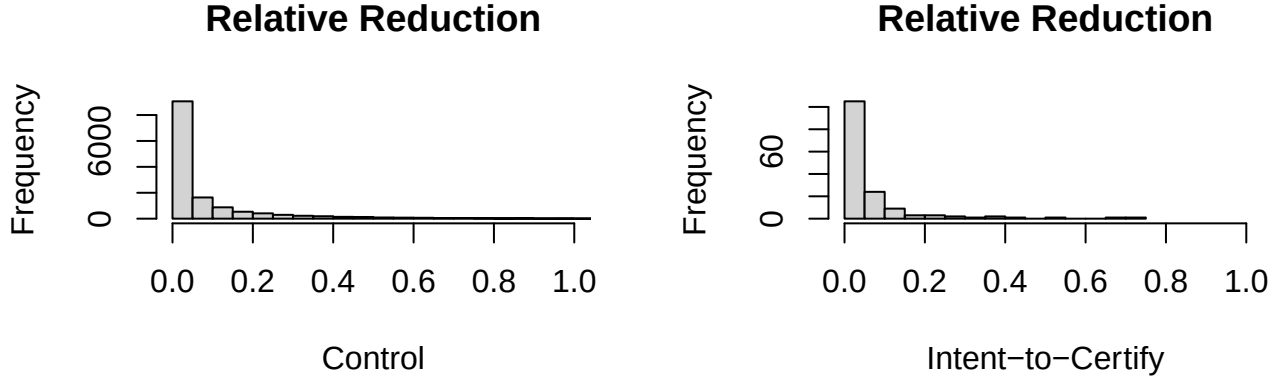
I compute each plantation-year's relative reduction, measured as

$$\text{reduct} = \frac{\text{f90for}(y) - \text{f90for}(y+1)}{\text{f90for}(y)}$$

, and plot histograms for both the intent-to-certify group and the 'control' group.

```
df$cert_next <- df$certintent_year == (df$year + 1)
reduct_df <- df
reduct_df <- filter(reduct_df, year <= 2014 & year >= 2008)
reduct_df$tf <- log(reduct_df$reduct +
                    1/2*min(reduct_df$reduct[reduct_df$reduct > 0]))
reduct_df$rc <- reduct_df$reduct/reduct_df$f90_for
par(mfrow=c(1,2))
hist(reduct_df$rc[reduct_df$cert_next==0],
     main="Relative Reduction", xlab="Control", breaks=20,
     xlim=c(0,1))
hist(reduct_df$rc[reduct_df$cert_next==1],
     main="Relative Reduction", xlab="Intent-to-Certify",
```

```
breaks=20, xlim=c(0,1))
```



The histograms appear quite similar to one another, demonstrating that the intent to certify may not reduce deforestation. However, the intent-to-certify group may have a slightly lower relative reduction in dense forest cover, as depicted by its maximum relative change being less than 0.8. Let's create a variable, called `reduct`, that is defined as `reduct = f90for(y) - f90for(y+1)`. We model each reduction as a function of the other covariates and the variable `cert_next`. We should limit our analysis to the years during which it was possible to initiate certification the following year (2008-2014). In **Q2**, we fit a logistic regression model for `cert_next`. In doing so, we determine that most of the covariates are independent from `cert_next`. Excluding these covariates from any model will not bias our estimated coefficient corresponding to `cert_next`. Therefore, we only need to consider the `year`, `f90_for`, `f30_ptcl`, and `pp_2010` covariates and the covariates excluded from modeling (`id`, `district`, `province`, `island`). We should also limit our dataset to only include observations with `f90_for > 0`. Otherwise, we include observations in which there is no potential for reduction in `f90_for`. To test our hypothesis, we fit a linear mixed effects model with the transformed `reduction` variable as our response. Anticipating correlation between plantation-year observations from the same plantation, district, province, and even island, I include random effects for each geographic level. For plantation i in year j , our model is:

$$\log(\alpha + \text{reduction}_{ij}) = \beta_{\text{cert.intent.now}} \text{cert.intent.now}_{ij} + X\beta + \sigma_0 u_i + \sigma_1 v_{\text{district}} + \sigma_2 w_{\text{island}} + \sigma_3 s_{\text{province}} + \sigma_4 \epsilon_{ij};$$

$$u_i, v_{\text{district}}, w_{\text{island}}, s_{\text{province}}, \epsilon_{ij} \stackrel{iid}{\sim} N(0, 1).$$

We test the null, $H_0 : \beta_{\text{certintent.now}} = 0$ against the alternative: $\beta_{\text{certintent.now}} < 0$.

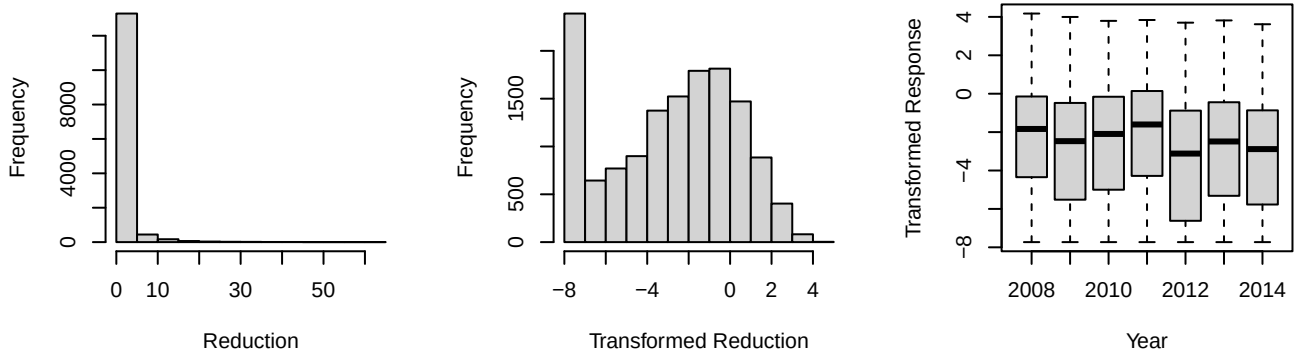
Our findings suggest that the intent-to-certify year does not have a significant effect on the reduction in deforestation.

```
reduct_df <- reduct_df %>% filter(f90_for > 0)
reduct_df$fdistrict <- factor(reduct_df$island)
reduct_df$fid <- factor(reduct_df$id)
fit <- lmer(tf~cert_next + factor(year) +
            f90_for + f30_ptcl + pp2010 + (1 | fid), data=reduct_df)
fit1 <- lmer(tf~cert_next + factor(year) +
            f90_for + f30_ptcl + pp2010 + (1 | fid) + (1 | district),
            data=reduct_df)
fit2 <- lmer(tf~cert_next + factor(year) +
            f90_for + f30_ptcl + pp2010 + (1 | fid) + (1 | district)
            + (1 | province), data=reduct_df)
fit3 <- lmer(tf~cert_next + factor(year) +
            f90_for + f30_ptcl + pp2010 + (1 | fid) + (1 | district)
            + (1 | province) + (1 | island), data=reduct_df)
fit3_partial <- lmer(tf~cert_next + factor(year) +
```

```

      f90_for + f30_ptcl + pp2010 + (1 | fid) + (1 | district)
    + (1 | province) + (1 | island),
    data=reduct_df[reduct_df$reduct>0,])
fit3null <- lmer(tf~factor(year) + f90_for + f30_ptcl + pp2010 + (1 | fid)
  + (1 | district) + (1 | province) + (1 | island),
  data=reduct_df)
fit3partialnull <- lmer(tf~factor(year) +
  f90_for + f30_ptcl + pp2010 + (1 | fid)
  + (1 | district) + (1 | province) + (1 | island),
  data=reduct_df[reduct_df$reduct>0,])
cmp <- anova(fit3, fit3null)
cmp_partial <- anova(fit3_partial, fit3partialnull)
beta <- summary(fit3)$coefficients[2,1]
betapartial <- summary(fit3_partial)$coefficients[2,1]
par(mfrow=c(1,3))
hist(reduct_df$reduct, main="", xlab="Reduction")
hist(reduct_df$tf, main="", xlab="Transformed Reduction")
plot(factor(reduct_df$year), reduct_df$tf, xlab="Year", ylab="Transformed Response")

```



It may make sense to include a time component in our covariance structure. It seems reasonable for the variance of reduction to decrease as less forestry is available to destroy. However, the plot above (on the right) demonstrates that after transforming the response, the variance is constant with respect to year.

Our estimated coefficients for the full and reduced data are $\hat{\beta}_{cert.intent.now} = -0.0579$, $\hat{\beta}_{certintent.now} = -0.1784$. On the full dataset, a likelihood ratio test yields a test statistic of 0.2103 on 1 degree of freedom. On the dataset restricted to positive `reduct` values, a likelihood ratio test yields a test statistic of 1.4744 on 1 degree of freedom, suggesting no evidence for a negative effect of intent-to-certify ($\hat{p} = 0.6465, 0.2246$ respectively).

Let $Z \in \mathbb{R}^{N \times K}$ such that $Z_{ik} = 1$ if $z_i = k$ and 0 otherwise. Let $Y \in \mathbb{R}^{N \times T}$ such that $Y_{it} = y_i^{(t)}$. Let $\Theta \in \mathbb{R}^{K \times T}$ such that $\Theta_{kt} = \hat{\mu}_k^{(t)}$. To compute the E step, we first compute the complete log-likelihood $P(Y, \Theta, Z)$. Note that the x_i^{area} values are given. For simplicity, let c_i denote the constant x_i^{area} . Note that

$$\begin{aligned}
P(Y, \Theta, Z) &= P(Z)P(\Theta | Z)P(Y | \Theta, Z) \\
&= P(Z)P(\Theta)P(Y | \Theta, Z) = \prod_{i=1}^N P(z_i = k) \prod_{t=1}^T \prod_{k=1}^K P(\mu_k^{(t)}) \prod_{t=1}^T \prod_{i=1}^N P(y_i^{(t)} | z_i, \mu_{z_i}^{(t)}) \\
&= \prod_{i=1}^N \prod_{k=1}^K \left[\frac{1}{K} \right]^{Z_{ik}} \prod_{t=1}^T \prod_{k=1}^K \Gamma(\mu_k^{(t)}; a_0, b_0) \prod_{t=1}^T \prod_{i=1}^N \prod_{k=1}^K \text{Pois}(y_i^{(t)}; c_i \mu_k^{(t)})^{Z_{ik}} \\
&= \prod_{i=1}^N \prod_{k=1}^K \left[\frac{1}{K} \right]^{Z_{ik}} \prod_{t=1}^T \prod_{k=1}^K \frac{b_0^{a_0}}{\Gamma(a_0)} \mu_k^{(t)(a_0-1)} e^{-b_0 \mu_k^{(t)}} \prod_{t=1}^T \prod_{i=1}^N \prod_{k=1}^K \left[e^{-c_i \mu_k^{(t)}} (c_i \mu_k^{(t)})^{y_i^{(t)}} \frac{1}{y_i^{(t)}!} \right]^{Z_{ik}} \\
&= \prod_{i=1}^N \prod_{k=1}^K \left[\frac{1}{K} \right]^{Z_{ik}} \prod_{t=1}^T \prod_{k=1}^K \frac{b_0^{a_0}}{\Gamma(a_0)} \Theta_{kt}^{(a_0-1)} e^{-b_0 \Theta_{kt}} \prod_{t=1}^T \prod_{i=1}^N \prod_{k=1}^K \left[e^{-c_i \Theta_{kt}} (c_i \Theta_{kt})^{Y_{it}} \frac{1}{Y_{it}!} \right]^{Z_{ik}},
\end{aligned}$$

where in the last line we replace the given notation with the matrix notation described above. The log complete likelihood gives

$$\begin{aligned}
\log(P(Y, \Theta, Z)) &= \sum_{i=1}^N \sum_{k=1}^K Z_{ik} [-\log(K)] + \sum_{t=1}^T \sum_{k=1}^K a_0 \log(b_0) - \log(\Gamma(a_0)) + (a_0 - 1) \log(\Theta_{kt}) - b_0 \Theta_{kt} \\
&\quad + \sum_{t=1}^T \sum_{i=1}^N \sum_{k=1}^K Z_{ik} [Y_{it} \log(c_i \Theta_{kt}) - c_i \Theta_{kt} - \log(Y_{it}!)].
\end{aligned}$$

In the E step, We take the expectation, conditional on Y and treating Θ as fixed.

$$E_{\Theta}[\log(P(Y, \Theta, Z)) | Y] =$$

$$\begin{aligned}
&= E_{\Theta} \left[\sum_{i=1}^N \sum_{k=1}^K Z_{ik} [-\log(K)] + \sum_{t=1}^T \sum_{k=1}^K a_0 \log(b_0) - \log(\Gamma(a_0)) + (a_0 - 1) \log(\Theta_{kt}) - b_0 \Theta_{kt} \right. \\
&\quad \left. + \sum_{t=1}^T \sum_{i=1}^N \sum_{k=1}^K Z_{ik} [Y_{it} \log(c_i \Theta_{kt}) - c_i \Theta_{kt} - \log(Y_{it}!)] \mid Y \right] \\
&= \sum_{i=1}^N \sum_{k=1}^K E_{\Theta}[Z_{ik} | Y] [-\log(K)] + \sum_{t=1}^T \sum_{k=1}^K a_0 \log(b_0) - \log(\Gamma(a_0)) + (a_0 - 1) \log(\Theta_{kt}) - b_0 \Theta_{kt} \\
&\quad + \sum_{t=1}^T \sum_{i=1}^N \sum_{k=1}^K E_{\Theta}[Z_{ik} | Y] [Y_{it} \log(c_i \Theta_{kt}) - c_i \Theta_{kt} - \log(Y_{it}!)].
\end{aligned}$$

It is evident that in the E-step, we must compute the quantity $E_\theta[Z_{ik} | Y]$. Note that

$$\begin{aligned}
E_\theta[Z_{ik} | Y] &= P(Z_{ik} = 1 | Y) \\
&= \frac{P(Y | Z_{ik} = 1)P(Z_{ik=1})}{\sum_{j=1}^K P(Y | Z_{ik} = 1)P(Z_{ij} = 1)} \\
&= \frac{\prod_{t=1}^T \text{Pois}(Y_{it}; c_i \Theta_{kt}) \frac{1}{K}}{\sum_{j=1}^K \prod_{t=1}^T \text{Pois}(Y_{it}; c_i \Theta_{jt}) \frac{1}{K}} \\
&= \frac{\prod_{t=1}^T \text{Pois}(Y_{it}; c_i \Theta_{kt})}{\sum_{j=1}^K \prod_{t=1}^T \text{Pois}(Y_{it}; c_i \Theta_{jt})}.
\end{aligned}$$

let $\tau_{ik} \equiv E_\theta[Z_{ik} | Y]$. Then $\tau \in \mathbb{R}^{N \times K}$ is a matrix. For maximum a posteriori estimation, we maximize the ELBO, plugging in a fixed τ .

We note that conditional on τ_{ik} and Y_{ik} , by Gamma-Poisson conjugacy, $\mu_k^{(t)}$ is Gamma-distributed in its posterior with parameters $a_0 + \sum_{i=1}^N \tau_{ik} Y_{ik}$, $b_0 + \sum_{i=1}^N \tau_{ik} c_i$. Then a posteriori, the maximum is the mode of the Gamma, which occurs at $\frac{a_0 + \sum_{i=1}^N \tau_{ik} Y_{it} - 1}{b_0 + \sum_{i=1}^N \tau_{ik} c_i}$. Alternatively, this expression may be derived by maximizing the ELBO $E_\Theta[\log(P(Y, \Theta, Z)) | Y]$ with respect to each Θ_{kt} . At the maximum,

$$\begin{aligned}
0 &= \frac{\partial E_\Theta[\log(P(Y, \Theta, Z)) | Y]}{\partial \Theta_{kt}} \\
&= \frac{a_0 - 1}{\Theta_{kt}} - b_0 + \sum_{i=1}^N \frac{\tau_{ik} Y_{it}}{\Theta_{kt}} - \sum_{i=1}^N \tau_{ik} c_i \\
\implies \Theta_{kt} &= \frac{a_0 + \sum_{i=1}^N \tau_{ik} Y_{it} - 1}{b_0 + \sum_{i=1}^N \tau_{ik} c_i}.
\end{aligned}$$

Therefore, in the M step, we set

$$\hat{\mu}_k^{(t)} = \frac{a_0 + \sum_{i=1}^N \tau_{ik} Y_{it} - 1}{b_0 + \sum_{i=1}^N \tau_{ik} c_i}.$$

Implementation is relatively straightforward: we alternate between updating $\{\tau_{ik}\}$ and updating $\{\hat{\mu}_k^{(t)}\}$ until convergence.